

# **VTU 3rd Semester - Operating Systems**

## **Module 1 - Internal Exam Notes (Theory)**

### **1. Introduction to Operating Systems**

An Operating System (OS) is a software that acts as an interface between the user and the computer hardware. It manages hardware resources and provides services for computer programs. Examples include Windows, Linux, and macOS.

### **2. Functions of an Operating System**

- Process Management – Creation, scheduling, and termination of processes.
- Memory Management – Allocation and deallocation of memory to processes.
- File System Management – Creation, deletion, and access of files.
- Device Management – Managing input/output devices.
- Security and Protection – Ensuring authorized access to resources.
- User Interface – Providing CLI or GUI to interact with users.

### **3. Operating System Services**

- Program Execution – Load and run programs.
- I/O Operations – Read/write operations on devices.
- File System Manipulation – Create, delete, read, write files.
- Communication Services – IPC between processes.
- Error Detection – Detect and handle errors.

### **4. System Calls**

System calls are functions provided by the OS that allow user programs to request services from the kernel. Examples include `fork()`, `exec()`, `read()`, `write()`, `open()`, `close()`.

### **5. Types of Operating Systems**

- Batch Operating System – Executes jobs in batches, no interaction during execution.
- Time-Sharing / Multi-User OS – Allows multiple users to share system resources concurrently.
- Distributed OS – Manages a group of independent computers as a single system.
- Network OS – Provides services over a network for multiple computers.
- Real-Time OS – Provides immediate processing for real-time applications.

### **6. System Structures**

Operating systems can be organized in different architectures:

- Monolithic – All OS services run in kernel space.
- Layered – OS divided into layers with specific functionalities.
- Microkernel – Minimal kernel, most services run in user space.
- Modular – OS built with separate modules that can be dynamically loaded.

### **7. Virtual Machines**

A Virtual Machine (VM) is an emulation of a physical computer that runs an operating system. It allows users to run multiple OSs on a single physical machine safely. Advantages: - Isolation from host OS - Easy testing and experimentation - Snapshots and backups

### 8. Example Internal Questions & Answers

| Question                | Short Answer (2-3 marks)                      | Long Answer (10 marks)                                          |
|-------------------------|-----------------------------------------------|-----------------------------------------------------------------|
| Define Operating System | Software interface between user and hardware. | An Operating System (OS) is system software that manages        |
| List OS functions       | Process, memory, file, device management.     | The main functions of an OS include: Process Management         |
| What are system calls?  | Functions to request OS services.             | System calls are functions provided by the operating system     |
| Different types of OS   | Batch, time-sharing, distributed, real-time.  | Operating systems can be classified as: Batch OS (jobs executed |

**End of Notes**